Appendices

A Beam Material Model

PROOF OF NEO HOOKEAN IN S

$$
\begin{aligned}
\boldsymbol{S} &= \mathrm{J}\boldsymbol{F^{-1}}\left[\frac{\mu}{J}\left(\boldsymbol{b}-\boldsymbol{I}\right)+\frac{\Lambda}{J}\ln J\boldsymbol{I}\right]\boldsymbol{F^{-T}} \\
&= \mathrm{J}\left[\frac{\mu}{\mathrm{J}}\left(\boldsymbol{F^{-1}b}-\boldsymbol{F^{-1}}\right)+\frac{\Lambda}{\mathrm{J}}\ln \mathrm{J}\boldsymbol{F^{-1}}\right]\boldsymbol{F^{-T}} \\
&= \mathrm{J}\left[\frac{\mu}{\mathrm{J}}\left(\boldsymbol{F^{-1}}\left(\boldsymbol{F}\boldsymbol{F^{T}}\right)-\boldsymbol{F^{-1}}\right)+\frac{\Lambda}{\mathrm{J}}\ln \mathrm{J}\boldsymbol{F^{-1}}\right]\boldsymbol{F^{-T}} \\
&= \mathrm{J}\left[\frac{\mu}{\mathrm{J}}\left(\boldsymbol{F^{T}}-\boldsymbol{F^{-1}}\right)\boldsymbol{F^{-T}}+\frac{\Lambda}{\mathrm{J}}\ln \mathrm{J}\boldsymbol{F^{-1}}\boldsymbol{F^{-T}}\right] \\
&= \mu\left(\boldsymbol{F^{T}}\boldsymbol{F^{-T}}-\left(\boldsymbol{F^{T}}\boldsymbol{F}\right)^{-1}\right)+\Lambda\ln \mathrm{J}\left(\boldsymbol{F^{T}}\boldsymbol{F}\right)^{-1}
\end{aligned}
$$

$$\therefore \boldsymbol{S} = \mu\left(\boldsymbol{I}-\boldsymbol{C^{-1}}\right)+\Lambda\ln \mathrm{J}\boldsymbol{C^{-1}} \tag{1}$$

PROOF OF C USING S Assuming Cartesian basis, $\boldsymbol{C} = \mathrm{C}^{\mathrm{i}}_{.\mathrm{j}}\boldsymbol{e}_{\mathrm{i}}\otimes\boldsymbol{e}^{\mathrm{j}}$ and $\boldsymbol{C}^{-1} = \left(\mathrm{C}^{-1}\right)^{\mathrm{i}}_{.\mathrm{j}}\boldsymbol{e}_{\mathrm{i}}\otimes\boldsymbol{e}^{\mathrm{j}}$

$$\mathrm{S}^{\mathrm{i}}_{.\mathrm{j}} = \mu\left[\delta^{\mathrm{i}}_{.\mathrm{i}}-\left(\mathrm{C}^{-1}\right)^{\mathrm{i}}_{.\mathrm{j}}\right]+\Lambda\ln \mathrm{J}\left(\mathrm{C}^{-1}\right)$$

$$\Rightarrow 2\frac{\partial \mathrm{S}^{\mathrm{i}}_{.\mathrm{j}}}{\partial \mathrm{C}^{\mathrm{a}}_{.\mathrm{b}}} = 2\mu\left[0-\frac{\partial\left(\mathrm{C}^{-1}\right)^{\mathrm{i}}_{.\mathrm{j}}}{\partial \mathrm{C}^{\mathrm{a}}_{.\mathrm{b}}}\right]+2\Lambda\frac{\partial\left(\ln \mathrm{J}\left(\mathrm{C}^{-1}\right)^{\mathrm{i}}_{.\mathrm{j}}\right)}{\partial \mathrm{C}^{\mathrm{a}}_{.\mathrm{b}}}$$

Using EQUATION 15 FROM DOCUMENT

$$
\begin{aligned}
2\frac{\partial \mathrm{S}^{\mathrm{i}}_{.\mathrm{j}}}{\partial \mathrm{C}^{\mathrm{a}}_{.\mathrm{b}}} &= 2\mu\left[0-\frac{\partial\left(\mathrm{C}^{-1}\right)^{\mathrm{i}}_{.\mathrm{j}}}{\partial \mathrm{C}^{\mathrm{a}}_{.\mathrm{b}}}\right]+2\Lambda\left(\mathrm{C}^{-1}\right)^{\mathrm{i}}_{.\mathrm{j}}\boldsymbol{e}_{\mathrm{i}}\otimes\boldsymbol{e}^{\mathrm{j}}\otimes\frac{\partial\left(\ln \mathrm{J}\right)}{\partial \mathrm{C}^{\mathrm{a}}_{.\mathrm{b}}}+2\Lambda\ln \mathrm{J}\frac{\partial\left(\mathrm{C}^{-1}\right)^{\mathrm{i}}_{.\mathrm{j}}}{\partial \mathrm{C}^{\mathrm{a}}_{.\mathrm{b}}} \\
&= 2\left(-\mu+\Lambda\ln \mathrm{J}\right)\frac{\partial\left(\mathrm{C}^{-1}\right)^{\mathrm{i}}_{.\mathrm{j}}}{\partial \mathrm{C}^{\mathrm{a}}_{.\mathrm{b}}}+2\Lambda\left(\mathrm{C}^{-1}\right)^{\mathrm{i}}_{.\mathrm{j}}\boldsymbol{e}_{\mathrm{i}}\otimes\boldsymbol{e}^{\mathrm{j}}\otimes\left[\frac{1}{\mathrm{J}}\frac{\partial \mathrm{J}}{\partial \mathrm{C}^{\mathrm{a}}_{.\mathrm{b}}}\right]
\end{aligned}
$$

Using WRITE REFERENCES FROM APPENDIX,

$$
\begin{aligned}
2\frac{\partial \boldsymbol{S}}{\partial \boldsymbol{C}} = 2\left(-\mu+\Lambda\ln \mathrm{J}\right)&\left(\mathrm{C}^{-1}\right)^{\mathrm{i.a}}\left(\mathrm{C}^{-1}\right)^{\mathrm{b.j}}\boldsymbol{e}_{\mathrm{i}}\otimes\boldsymbol{e}^{\mathrm{a}}\otimes\boldsymbol{e}_{\mathrm{b}}\otimes\boldsymbol{e}^{\mathrm{j}} \\
&+\Lambda\left(\mathrm{C}^{-1}\right)^{\mathrm{i}}_{.\mathrm{j}}\left(\mathrm{C}^{-1}\right)^{\mathrm{a}}_{.\mathrm{b}}\boldsymbol{e}_{\mathrm{i}}\otimes\boldsymbol{e}^{\mathrm{j}}\otimes\boldsymbol{e}_{\mathrm{a}}\otimes\boldsymbol{e}^{\mathrm{b}}
\end{aligned}
$$

$$\because \mathbb{C} = 2\frac{\partial \boldsymbol{S}}{\partial \boldsymbol{C}}$$

$$\mathbb{C}^{\mathrm{i.a.}}_{.\mathrm{j.b}} = 2\left(-\mu+\Lambda\ln \mathrm{J}\right)\left(\mathrm{C}^{-1}\right)^{\mathrm{i}}_{.\mathrm{a}}\left(\mathrm{C}^{-1}\right)^{\mathrm{b}}_{.\mathrm{j}}+\Lambda\left(\mathrm{C}^{-1}\right)^{\mathrm{i}}_{.\mathrm{j}}\left(\mathrm{C}^{-1}\right)^{\mathrm{a}}_{.\mathrm{b}} \tag{2}$$

YOOO PUSH FORWARD PROOF

$$\mathbb{c} = 2\left(-\mu+\Lambda\ln \mathrm{J}\right)\left(\mathrm{C}^{-1}\right)^{\mathrm{i}}_{.\mathrm{a}}\left(\mathrm{C}^{-1}\right)^{\mathrm{b}}_{.\mathrm{j}}\boldsymbol{F}\boldsymbol{e}_{\mathrm{i}}\otimes\boldsymbol{e}^{\mathrm{a}}\boldsymbol{F^{T}}\otimes\boldsymbol{F}\boldsymbol{e}_{\mathrm{b}}\otimes\boldsymbol{e}^{\mathrm{j}}\boldsymbol{F^{T}}+\Lambda\left(\mathrm{C}^{-1}\right)^{\mathrm{i}}_{.\mathrm{j}}\left(\mathrm{C}^{-1}\right)^{\mathrm{a}}_{.\mathrm{b}}\boldsymbol{F}\boldsymbol{e}_{\mathrm{i}}\otimes\boldsymbol{e}^{\mathrm{j}}\boldsymbol{F^{T}}\otimes\boldsymbol{F}\boldsymbol{e}_{\mathrm{a}}\otimes\boldsymbol{e}^{\mathrm{b}}\boldsymbol{F^{T}}$$

Let

$$\mathbb{I} = 2\left(-\mu+\Lambda\ln J\right)\left(C^{-1}\right)^{i}_{.a}\left(C^{-1}\right)^{b}_{.j}\boldsymbol{F}\boldsymbol{e}_{i}\otimes\boldsymbol{e}^{a}\boldsymbol{F^{T}}\otimes\boldsymbol{F}\boldsymbol{e}_{b}\otimes\boldsymbol{e}^{j}\boldsymbol{F^{T}} \tag{3}$$

$$\mathbb{III} = \Lambda\left(C^{-1}\right)^{i}_{.j}\left(C^{-1}\right)^{a}_{.b}\boldsymbol{F}\boldsymbol{e}_{i}\otimes\boldsymbol{e}^{j}\boldsymbol{F^{T}}\otimes\boldsymbol{F}\boldsymbol{e}_{a}\otimes\boldsymbol{e}^{b}\boldsymbol{F^{T}} \tag{4}$$

$$\mathbb{I} = 2\left(-\mu + \Lambda \ln J\right)\left(C^{-1}\right)^{i}_{.a}\left(C^{-1}\right)^{b}_{.j}\left(F^{p}_{.q}\boldsymbol{e}_p \otimes \boldsymbol{e}^q.\boldsymbol{e}_i\right) \otimes \left(\boldsymbol{e}^a.\boldsymbol{e}_r \otimes \boldsymbol{e}^s\left(F^T\right)^{v}_{.w}\right) \otimes \left(F^{t}_{.u}\boldsymbol{e}_t \otimes \boldsymbol{e}^u.\boldsymbol{e}_b\right) \otimes \left(\boldsymbol{e}^j.\boldsymbol{e}_v \otimes \boldsymbol{e}^w\left(F^T\right)^{v}_{.w}\right)$$

$$\begin{aligned}
\mathbb{I} &= 2\left(-\mu + \Lambda \ln J\right)\left(F^{-1}F^{-T}\right)^{i}_{.a}\left(F^{-1}F^{-T}\right)^{b}_{.j}\left[F^{p}_{.q}\delta^{q}_{.i}\delta^{a}_{.r}(F^T)^{r}_{.s}F^{t}_{.u}\delta^{u}_{.b}\delta^{j}_{.v}(F^T)^{v}_{.w}\right]\boldsymbol{e}_p \otimes \boldsymbol{e}^s \otimes \boldsymbol{e}_t \otimes \boldsymbol{e}^w \\
&= 2\left(-\mu + \Lambda \ln J\right)\left(F^{-1}\right)^{i}_{.c}\left(F^{-T}\right)^{c}_{.a}\left(F^{-1}\right)^{b}_{.d}\left(F^{-T}\right)^{d}_{.j}\left[F^{p}_{.i}\delta^{a}_{.r}(F^T)^{r}_{.s}F^{t}_{.b}\delta^{j}_{.v}(F^T)^{v}_{.w}\right]\boldsymbol{e}_p \otimes \boldsymbol{e}^s \otimes \boldsymbol{e}_t \otimes \boldsymbol{e}^w \\
&= 2\left(-\mu + \Lambda \ln J\right)\left[F^{p}_{.i}(F^{-1})^{i}_{.c}F^{t}_{.b}(F^{-1})^{b}_{.d}\right]\left[(F^{-T})^{c}_{.a}\delta^{a}_{.r}(F^T)^{r}_{.s}\right]\left[(F^{-T})^{d}_{.j}\delta^{j}_{.v}(F^T)^{v}_{.w}\right]\boldsymbol{e}_p \otimes \boldsymbol{e}^s \otimes \boldsymbol{e}_t \otimes \boldsymbol{e}^w \\
&= 2\left(-\mu + \Lambda \ln J\right)\delta^{p}_{.c}\delta^{t}_{.d}(F^{-T})^{c}_{.r}(F^T)^{r}_{.s}(F^{-T})^{d}_{.v}(F^T)^{v}_{.w}\boldsymbol{e}_p \otimes \boldsymbol{e}^s \otimes \boldsymbol{e}_t \otimes \boldsymbol{e}^w \\
&= 2\left(-\mu + \Lambda \ln J\right)\delta^{p}_{.c}\delta^{t}_{.d}\delta^{c}_{.s}\delta^{d}_{.w}\boldsymbol{e}_p \otimes \boldsymbol{e}^s \otimes \boldsymbol{e}_t \otimes \boldsymbol{e}^w
\end{aligned}$$

$$\mathbb{I} = 2\left(-\mu + \Lambda \ln J\right)\delta^{p}_{.s}\delta^{t}_{.w}\boldsymbol{e}_p \otimes \boldsymbol{e}^s \otimes \boldsymbol{e}_t \otimes \boldsymbol{e}^w \tag{5}$$

Similarly

$$\mathbb{III} = \frac{\Lambda}{2}\left(C^{-1}\right)^{i}_{.j}\left(C^{-1}\right)^{a}_{.b}\left(F^{p}_{.q}\boldsymbol{e}_p \otimes \boldsymbol{e}^q.\boldsymbol{e}_i\right) \otimes \left(\boldsymbol{e}^j.\boldsymbol{e}_v \otimes \boldsymbol{e}^w\left(F^T\right)^{v}_{.w}\right) \otimes \left(F^{r}_{.s}\boldsymbol{e}_r \otimes \boldsymbol{e}^s.\boldsymbol{e}_a\right) \otimes \left(\boldsymbol{e}^b.\boldsymbol{e}_t \otimes \boldsymbol{e}^u\left(F^T\right)^{t}_{.u}\right)$$

$$\begin{aligned}
\mathbb{III} &= \frac{\Lambda}{2}\left(C^{-1}\right)^{i}_{.j}\left(C^{-1}\right)^{a}_{.b}\left[F^{p}_{.q}\delta^{q}_{.i}\delta^{j}_{.v}(F^T)^{v}_{.w}F^{r}_{.s}\delta^{s}_{.a}\delta^{b}_{.t}(F^T)^{t}_{.u}\right]\boldsymbol{e}_p \otimes \boldsymbol{e}^w \otimes \boldsymbol{e}_r \otimes \boldsymbol{e}^u \\
&= \frac{\Lambda}{2}\left(F^{-1}F^{-T}\right)^{i}_{.j}\left(F^{-1}F^{-T}\right)^{a}_{.b}\left[F^{p}_{.i}\delta^{j}_{.v}(F^T)^{v}_{.w}F^{r}_{.a}\delta^{b}_{.t}(F^T)^{t}_{.u}\right]\boldsymbol{e}_p \otimes \boldsymbol{e}^w \otimes \boldsymbol{e}_r \otimes \boldsymbol{e}^u \\
&= \frac{\Lambda}{2}\left(F^{-1}\right)^{i}_{.c}\left(F^{-T}\right)^{c}_{.j}\left(F^{-1}\right)^{a}_{.d}\left(F^{-T}\right)^{d}_{.b}\left[F^{p}_{.i}\delta^{j}_{.v}(F^T)^{v}_{.w}F^{r}_{.a}\delta^{b}_{.t}(F^T)^{t}_{.u}\right]\boldsymbol{e}_p \otimes \boldsymbol{e}^w \otimes \boldsymbol{e}_r \otimes \boldsymbol{e}^u \\
&= \frac{\Lambda}{2}(F^{p}_{.i}(F^{-1})^{i}_{.c})(F^{r}_{.a}(F^{-1})^{a}_{.d})\left[\left(F^{-T}\right)^{c}_{.j}\delta^{j}_{.v}(F^T)^{v}_{.w}\right]\left[\left(F^{-T}\right)^{d}_{.b}\delta^{b}_{.t}(F^T)^{t}_{.u}\right]\boldsymbol{e}_p \otimes \boldsymbol{e}^w \otimes \boldsymbol{e}_r \otimes \boldsymbol{e}^u \\
&= \frac{\Lambda}{2}\delta^{p}_{.c}\delta^{r}_{.d}\left[(F^{-T})^{c}_{.v}(F^T)^{v}_{.w}\right]\left[(F^{-T})^{d}_{.t}(F^T)^{t}_{.u}\right]\boldsymbol{e}_p \otimes \boldsymbol{e}^w \otimes \boldsymbol{e}_r \otimes \boldsymbol{e}^u \\
&= \frac{\Lambda}{2}\delta^{p}_{.c}\delta^{r}_{.d}\delta^{c}_{.w}\delta^{d}_{.u}\boldsymbol{e}_p \otimes \boldsymbol{e}^w \otimes \boldsymbol{e}_r \otimes \boldsymbol{e}^u \\
&= \frac{\Lambda}{2}\delta^{p}_{.w}\delta^{r}_{.u}\boldsymbol{e}_p \otimes \boldsymbol{e}^w \otimes \boldsymbol{e}_r \otimes \boldsymbol{e}^u
\end{aligned}$$

As all indices are dummy indices, they can be replaced as following

$$\mathbb{III} = \frac{\Lambda}{2}\delta^{p}_{.w}\delta^{s}_{.t}\boldsymbol{e}_p \otimes \boldsymbol{e}^w \otimes \boldsymbol{e}_s \otimes \boldsymbol{e}^t \tag{6}$$

B Shell Formulations

B.1 Weak Form Derivation

$$\int_{\mathcal{S}} \boldsymbol{w}\left[\boldsymbol{t}^{\alpha}_{;\alpha} + \boldsymbol{f}\right]\,\mathrm{d}a = 0. \tag{7}$$

Decomposing $\boldsymbol{w}$ into its surface and normal components

$$0 = \int_{\mathcal{S}} \left[w_\gamma \boldsymbol{a}^\gamma + w\boldsymbol{n}\right]\left[\sigma^{\beta\alpha}{}_{;\alpha}\boldsymbol{a}_\beta + \sigma^{\beta\alpha}b_{\beta\alpha}\boldsymbol{n} + \boldsymbol{f}\right]\,\mathrm{d}a \tag{8}$$

$$0 = \int_{\mathcal{S}} w_\gamma \boldsymbol{a}^\gamma\left[\sigma^{\beta\alpha}{}_{;\alpha}\boldsymbol{a}_\beta + f^\alpha \boldsymbol{a}_\alpha\right]\,\mathrm{d}a + \int_{\mathcal{S}} w\boldsymbol{n}\left[\sigma^{\beta\alpha}b_{\beta\alpha}\boldsymbol{n} + p\boldsymbol{n}\right]\,\mathrm{d}a \tag{9}$$

$$
\begin{aligned}
0 &= \int_{\mathcal{S}} \left(w_\gamma \sigma^{\beta\alpha}{}_{;\alpha} \delta^\gamma_\beta + f^\alpha \delta^\gamma_\alpha \right) \, \mathrm{d}a + \int_{\mathcal{S}} w \left[\sigma^{\beta\alpha} b_{\beta\alpha} + p \right] \, \mathrm{d}a \\
&= \int_{\mathcal{S}} \left[w_\beta \sigma^{\beta\alpha}{}_{;\alpha} + f^\gamma \right] \, \mathrm{d}a + \int_{\mathcal{S}} \left[w \sigma^{\beta\alpha} b_{\beta\alpha} + p \right] \, \mathrm{d}a \\
&= \int_{\mathcal{S}} \left[w_\beta \sigma^{\beta\alpha}{}_{;\alpha} + f^\gamma \right] \, \mathrm{d}a + \int_{\mathcal{S}} \left[w \sigma^{\beta\alpha} b_{\beta\alpha} + p \right] \, \mathrm{d}a \\
&= \int_{\mathcal{S}} \left[w_\gamma \sigma^{\gamma\alpha}{}_{;\alpha} + w_\gamma f^\gamma \right] \, \mathrm{d}a + \int_{\mathcal{S}} \left[w \sigma^{\beta\alpha} b_{\beta\alpha} + p \right] \, \mathrm{d}a
\end{aligned}
\tag{10}
$$

Using divergence theorem

$$
\begin{aligned}
\int_{\mathcal{S}} w_\gamma \sigma^{\gamma\alpha}{}_{;\alpha} \, \mathrm{d}a &= \int_{\mathcal{S}} \left(w_\gamma \sigma^{\gamma\alpha} \right)_{;\alpha} \, \mathrm{d}a - \int_{\mathcal{S}} w_{\gamma;\alpha} \sigma^{\gamma\alpha} \, \mathrm{d}a \\
&= \int_{\partial\mathcal{S}} w_\gamma \sigma^{\gamma\alpha} m_\alpha \, \mathrm{d}s - \int_{\mathcal{S}} w_{\gamma;\alpha} \sigma^{\gamma\alpha} \, \mathrm{d}a
\end{aligned}
\tag{11}
$$

using $\boldsymbol{t} = m_\alpha \boldsymbol{t}^\alpha = m_\alpha \sigma^{\beta\alpha} \boldsymbol{a}_\beta$ it is possible to rewrite the surfrace integral of (**??**) as

$$
\int_{\partial\mathcal{S}} w_\gamma \sigma^{\gamma\alpha} m_\alpha \, \mathrm{d}s = \int_{\partial\mathcal{S}} w_\gamma \bar{t}^\gamma \, \mathrm{d}s \tag{12}
$$

the weak form can be written as,

$$
0 = \int_{\partial\mathcal{S}} w_\gamma \bar{t}^\gamma \, \mathrm{d}s - \int_{\mathcal{S}} w_{\gamma;\alpha} \sigma^{\gamma\alpha} \, \mathrm{d}a + \int_{\mathcal{S}} w_\gamma f^\gamma \, \mathrm{d}a + \int_{\mathcal{S}} w \sigma^{\beta\alpha} b_{\beta\alpha} \, \mathrm{d}a + \int_{\mathcal{S}} w p \, \mathrm{d}s. \tag{13}
$$

B.2 Computation of $w_{\alpha;\beta}$

$$
\begin{aligned}
w_{\alpha;\beta} &= \left(\boldsymbol{w} \cdot \boldsymbol{a}_\alpha \right)_{;\beta} = \boldsymbol{w}_{;\beta} \cdot \boldsymbol{a}_\alpha + \boldsymbol{w} \cdot \boldsymbol{a}_{\alpha;\beta} \\
&= w_{;\beta} \boldsymbol{a}_\alpha + \left(w_\gamma \boldsymbol{a}^\gamma + w \boldsymbol{n} \right) a_{\alpha;\beta} \\
&= \boldsymbol{w}_{;\beta} \cdot \boldsymbol{a}_\alpha + w_\gamma \boldsymbol{a}^\gamma \cdot \boldsymbol{a}_{\alpha;\beta} + w \boldsymbol{n} \cdot \boldsymbol{a}_{\alpha;\beta} \\
&= \boldsymbol{w}_{;\beta} \cdot \boldsymbol{a}_\alpha + w b_{\alpha\beta}
\end{aligned}
\tag{14}
$$

B.3 Compliance Matrix

With the help of equations (**??**) and (**??**), the Kircchhof stress tensor can be written as

$$
\begin{aligned}
\tau^{\alpha\beta} &= 2 \frac{\partial W}{\partial a_{\alpha\beta}} = 2 \left[\frac{\Lambda}{4} \left(\frac{\partial J^2}{\partial a_{\alpha\beta}} - 2 \frac{\partial \ln J}{\partial a_{\alpha\beta}} \right) + \frac{\mu}{2} \left(\frac{\partial I_1}{\partial a_{\alpha\beta}} - 2 \frac{\partial \ln J}{\partial a_{\alpha\beta}} \right) \right] \\
&= \frac{\Lambda}{2} \left(\frac{\partial J^2}{\partial J} \frac{\partial J}{\partial a_{\alpha\beta}} - 2 \frac{\partial \ln J}{\partial J} \frac{\partial J}{\partial a_{\alpha\beta}} \right) + \mu \left(\frac{\partial I_1}{\partial a_{\alpha\beta}} - 2 \frac{\partial \ln J}{\partial J} \frac{\partial J}{\partial a_{\alpha\beta}} \right)
\end{aligned}
\tag{15}
$$

Computing the derivative of J as defined in (**??**) with respect to the metric tensor on the surface,

$$
\frac{\partial J}{\partial a_{\alpha\beta}} = \frac{1}{J_A} \frac{\partial J_a}{\partial a_{\alpha\beta}} = \frac{1}{J_A} \frac{\partial \sqrt{\det a_{\alpha\beta}}}{\partial \det a_{\alpha\beta}} \frac{\partial \det a_{\alpha\beta}}{\partial a_{\alpha\beta}} = \frac{1}{J_A} \frac{1}{2} \det a_{\alpha\beta}{}^{-\frac{1}{2}} \det a_{\alpha\beta} \left(a^{-1}_{\alpha\beta} \right)^T = \frac{1}{2 J_A} J_a a^{\alpha\beta} = \frac{J}{2} a^{\alpha\beta} \tag{16}
$$

and using (**??**) and (**??**), equation (**??**) can be written as

$$
\begin{aligned}
\tau^{\alpha\beta} &= \frac{\Lambda}{2} \left(2 J \frac{J}{2} a^{\alpha\beta} - \frac{1}{J} \frac{J}{2} a^{\alpha\beta} \right) + \mu \left(A^{\alpha\beta} - 2 \frac{1}{J} \frac{J}{2} a^{\alpha\beta} \right) \\
&= \frac{\Lambda}{2} \left(J^2 - 1 \right) a^{\alpha\beta} + \mu \left(A^{\alpha\beta} - a^{\alpha\beta} \right).
\end{aligned}
\tag{17}
$$

The compliance matrix in the current configuration is defined as

$$
c^{\alpha\beta\gamma\delta} = \frac{\partial^2 W}{\partial a_{\alpha\beta} \partial a_{\gamma\delta}} = 2 \frac{\partial \tau^{\alpha\beta}}{\partial a_{\gamma\delta}} \tag{18}
$$

Equation (**??**) can be rewritten as

$$\tau^{\alpha\beta} = \frac{\Lambda}{2}\left(J^2-1\right) + \mu A^{\alpha\beta} - \mu a^{\alpha\beta} = \left(\frac{\Lambda}{2}\left(J^2-1\right) - \mu\right) a^{\alpha\beta} - \mu A^{\alpha\beta} \tag{19}$$

With (**??**), equation (**??**) can be written as

$$\begin{aligned} c^{\alpha\beta\gamma\delta} &= 2\left[\frac{\partial}{\partial a_{\gamma\delta}}\left(\frac{\Lambda}{2}\left(J^2-1\right)-\mu\right)a^{\alpha\beta} + \left(\frac{\Lambda}{2}\left(J^2-1\right)-\mu\right)\frac{\partial a^{\alpha\beta}}{\partial a_{\gamma\delta}}\right] \\ &= 2\left[\frac{\Lambda}{2}\frac{\partial J^2}{\partial a_{\gamma\delta}}a^{\alpha\beta} + \left(\frac{\Lambda}{2}\left(J^2-1\right)-\mu\right)\frac{1}{2}m^{\alpha\beta\gamma\delta}\right] \\ &= \Lambda J^2 a^{\alpha\beta}a^{\gamma\delta} + \left(\frac{\Lambda}{2}\left(J^2-1\right)-\mu\right)m^{\alpha\beta\gamma\delta} \end{aligned} \tag{20}$$

B.4 Discretisation and Stuff

$$w_\alpha = \boldsymbol{w}\cdot\boldsymbol{a}_\alpha = \left(\mathbf{N}\mathbf{w}_e\right)^T\boldsymbol{a}_\alpha = \mathbf{w}_e^T\mathbf{N}^T\boldsymbol{a}_\alpha = \mathbf{w}_e^T\mathbf{N}^T\left(\mathbf{N}_{,\alpha}\mathbf{x}_e\right) \tag{21}$$

$$w \approx \boldsymbol{w}\cdot\boldsymbol{n} = \left(\mathbf{N}\mathbf{w}_e\right)^T\boldsymbol{n} = \mathbf{w}_e^T\mathbf{N}^T\boldsymbol{n} \tag{22}$$

Using

$$w_{;\alpha} \approx \mathbf{N}_{,\alpha}\mathbf{w}_e + \mathbf{N}\mathbf{w}_{e,\alpha} = \mathbf{N}_{,\alpha}\mathbf{w}_e \tag{23}$$

the following can be calculated

$$\begin{aligned} w_{\alpha;\beta} &= \boldsymbol{w}_{;\beta}\boldsymbol{a}_\alpha + wb_{\alpha\beta} \\ &\approx \left(\mathbf{N}\mathbf{w}_e\right)_{;\beta}\cdot\boldsymbol{a}_\alpha + w\boldsymbol{n}\mathbf{N}_{,\alpha\beta}\mathbf{x}_e \\ &= \left(\mathbf{N}_{,\beta}\mathbf{w}_e\right)\cdot\boldsymbol{a}_\alpha + w\boldsymbol{n}\cdot\mathbf{N}_{,\alpha\beta}\mathbf{x}_e \\ &= \left(\mathbf{N}_{,\beta}\mathbf{w}_e\right)\cdot\left(\mathbf{N}_{,\alpha}\mathbf{x}_e\right) + w\boldsymbol{n}\mathbf{N}_{,\alpha\beta}\mathbf{x}_e \\ &= \mathbf{w}_e^T\mathbf{N}_{,\beta}^T\mathbf{N}_{,\alpha}\mathbf{x}_e + \left(\boldsymbol{w}\cdot\boldsymbol{n}\right)\boldsymbol{n}\cdot\mathbf{N}_{,\alpha\beta}\mathbf{x}_e \\ &= \mathbf{w}_e^T\mathbf{N}_{,\beta}^T\mathbf{N}_{,\alpha}\mathbf{x}_e + \mathbf{w}_e^T\mathbf{N}^T\boldsymbol{n}\left(\mathbf{N}_{,\alpha\beta}\cdot\boldsymbol{n}\right)\mathbf{x}_e \\ &= \mathbf{w}_e^T\left[\mathbf{N}_{,\beta}^T\mathbf{N}_{,\alpha} + \mathbf{N}^T\left(\boldsymbol{n}\otimes\boldsymbol{n}\right)\mathbf{N}_{,\alpha\beta}\right]\mathbf{x}_e \end{aligned} \tag{24}$$

B.5 Linearisation

B.5.1 Linearisation of $\boldsymbol{a}_\alpha$a-a

$$\boldsymbol{a}_\alpha = \mathbf{N}_{,\alpha}\mathbf{x}_e \tag{25}$$

According to the definition of the tangent vector (**??**), the linearisation of the undiscretised form is

Linearising the parametric definition of the surface

$$\boldsymbol{x} = \boldsymbol{x}\left(\xi^1,\,\xi^2\right), \tag{26}$$

$$\implies \Delta\boldsymbol{x} = \frac{\partial\boldsymbol{x}}{\partial\xi^1}\Delta\xi^1 + \frac{\partial\boldsymbol{x}}{\partial\xi^2}\Delta\xi^2, \tag{27}$$

and using the definition of the tangent vector (**??**),

$$\boldsymbol{a}_\alpha = \boldsymbol{x}_{,\alpha} == \$frac\partial\partial\xi^\alpha\left(\boldsymbol{x}\left(\xi^1,\,\xi^2\right)\right), \tag{28}$$

the linearisation of $\boldsymbol{a}_\alpha$ can be written as

$$\Delta\boldsymbol{a}_\alpha = \Delta\left(\boldsymbol{x}_{,\alpha}\right) = \frac{\partial}{\xi^1}\left[\frac{\partial\boldsymbol{x}}{\partial\xi^\alpha}\right]\Delta\xi^1 + \frac{\partial}{\partial\xi^2}\left[\frac{\partial\boldsymbol{x}}{\partial\xi^\alpha}\right]\Delta\xi^2. \tag{29}$$

Using the property that the differential operator commutes and (**??**)

$$\Delta\boldsymbol{a}_\alpha = \frac{\partial}{\partial\xi^\alpha}\left[\frac{\partial\boldsymbol{x}}{\partial\xi^1}\Delta\xi^1 + \frac{\partial\boldsymbol{x}}{\partial\xi^2}\Delta\xi^2\right] = \frac{\partial}{\partial\xi^\alpha}\Delta\boldsymbol{x}. \tag{30}$$

Substituting the discretised value of $\Delta\boldsymbol{x}$ in (**??**),

$$\Delta\boldsymbol{a}_\alpha \approx \frac{\partial}{\partial\xi^\alpha}\left[\mathbf{N}\Delta\mathbf{x}_e\right] = \mathbf{N}_{,\alpha}\Delta\mathbf{x}_e \tag{31}$$

B.5.2 Linearisation of $a_{\alpha\beta}$

$$a_{\alpha\beta} = \boldsymbol{a}_\alpha \cdot \boldsymbol{a}_\beta \tag{32}$$

Using (**??**)

$$\begin{aligned}
\Delta a_{\alpha\beta} &= \Delta\left(\boldsymbol{a}_\alpha \cdot \boldsymbol{a}_\beta\right) = \Delta\boldsymbol{a}_\alpha \cdot \boldsymbol{a}_\beta + \boldsymbol{a}_\alpha \cdot \Delta\boldsymbol{a}_\beta = \boldsymbol{a}_\beta \cdot \Delta\boldsymbol{a}_\alpha + \boldsymbol{a}_\alpha \cdot \Delta\boldsymbol{a}_\beta \approx \left(\boldsymbol{a}_\beta \cdot \mathbf{N}_\alpha + \boldsymbol{a}_\alpha \cdot \mathbf{N}_\beta\right)\Delta\mathbf{x}_e \\
&= \left[\left(\mathbf{N}_{,\beta}\mathbf{x}_e\right) \cdot \mathbf{N}_{,\alpha}\right]\Delta\mathbf{x}_e + \left[\left(\mathbf{N}_\alpha\mathbf{x}_e\right) \cdot \mathbf{N}_\beta\right]\Delta\mathbf{x}_e = \left[\mathbf{x}_e^T\mathbf{N}_{,\beta}^T\mathbf{N}_\alpha + \mathbf{x}_e^T\mathbf{N}_{,\alpha}^T\mathbf{N}_{,\beta}\right]\Delta\mathbf{x}_e \\
&= \mathbf{x}_e^T\left[\mathbf{N}_{,\beta}^T\mathbf{N}_{,\alpha} + \mathbf{N}_\alpha^T\mathbf{N}_\beta\right]\Delta\mathbf{x}_e
\end{aligned} \tag{33}$$

B.5.3 Linearisation of J

Using the chain rule it can be written

$$\Delta J = \frac{\partial J}{\partial J_a}\frac{\partial J_a}{\partial a_{\gamma\delta}}\frac{\gamma\delta}{\partial a_\alpha}\Delta\boldsymbol{a}_\alpha. \tag{34}$$

Using the definitions in (**??**)

$$\begin{aligned}
\Delta J &= \frac{1}{J_A}\frac{\partial\sqrt{\det a_{\gamma\delta}}}{\partial a_{\gamma\delta}}\frac{\partial a_{\gamma\delta}}{\partial\boldsymbol{a}_\alpha}\Delta\boldsymbol{a}_\alpha \\
&= \left(\frac{1}{J_A}\right)\left(\frac{1}{2\sqrt{\det a_{\gamma\delta}}}\frac{\partial\left(\det a_{\gamma\delta}\right)}{\partial a_{\gamma\delta}}\right)\left(\frac{\partial a_{\gamma\delta}}{\partial\boldsymbol{a}_\alpha}\right)\Delta\boldsymbol{a}_\alpha \\
&= \left(\frac{1}{J_A}\right)\left(\frac{1}{2J_a}J_a^2 a^{\gamma\delta}\right)\left(\frac{\partial a_{\gamma\delta}}{\partial\boldsymbol{a}_\alpha}\right)\Delta\boldsymbol{a}_\alpha \\
&= \frac{J_a}{2J_A}a^{\gamma\delta}\frac{\partial\left(\boldsymbol{a}_\gamma \cdot \boldsymbol{a}_\delta\right)}{\partial\boldsymbol{a}_\alpha}\Delta\boldsymbol{a}_\alpha \\
&= \frac{J}{2}a^{\gamma\delta}\left[\delta^\alpha_{\cdot\gamma}\boldsymbol{a}_\delta + \boldsymbol{a}_\gamma\delta_\delta^{\cdot\alpha}\right]\Delta\boldsymbol{a}_\alpha \\
&= \frac{J}{2}\left[a^{\alpha\delta}\boldsymbol{a}_\delta + a^{\alpha\gamma}\boldsymbol{a}_\gamma\right]\Delta\boldsymbol{a}_\alpha \\
&= \frac{J}{2}\left(2\boldsymbol{a}^\alpha\right)\Delta\boldsymbol{a}_\alpha \\
\implies \Delta J &= J\boldsymbol{a}^\alpha\Delta\boldsymbol{a}_\alpha
\end{aligned} \tag{35}$$

With the discretised definitions (**??**) is given as

$$\Delta J \approx J\boldsymbol{a}^\alpha \cdot \mathbf{N}_{,\alpha}\Delta\mathbf{x}_e. \tag{36}$$

B.5.4 Linearisation of $a^{\alpha\beta}$

B.6 Something Else

$$\left(\boldsymbol{n} \otimes \boldsymbol{n}\right) = \tilde{\mathbf{1}} - \mathbf{1} \tag{37}$$

$$\begin{aligned}
\Rightarrow \left(\boldsymbol{n} \otimes \boldsymbol{n}\right)\boldsymbol{a}_{\alpha,\beta} &= \left(\tilde{\mathbf{1}} - \mathbf{1}\right)\boldsymbol{a}_{\alpha,\beta} \\
&= \tilde{\mathbf{1}}\boldsymbol{a}_{\alpha,\beta} - \left(\boldsymbol{a}_\gamma \otimes \boldsymbol{a}^\gamma\right)\boldsymbol{a}_{\alpha,\beta} \\
&= \boldsymbol{a}_{\alpha,\beta} - \left(\boldsymbol{a}_{\alpha,\beta} \cdot \boldsymbol{a}^\gamma\right)\boldsymbol{a}_\gamma \\
&= \boldsymbol{a}_{\alpha,\beta} - \Gamma^\gamma_{\alpha\beta}\boldsymbol{a}_\gamma \\
&= \boldsymbol{n} \cdot \boldsymbol{a}_{\alpha;\beta}
\end{aligned} \tag{38}$$

$$\begin{aligned}
\boldsymbol{n} \cdot \boldsymbol{a}_{\alpha;\beta} &= \boldsymbol{n} \cdot \left(\boldsymbol{n} \otimes \boldsymbol{n}\right)\boldsymbol{a}_{\alpha,\beta} \\
&= \left(\boldsymbol{n} \cdot \boldsymbol{n}\right)\left(\boldsymbol{a}_{\alpha,\beta} \cdot \boldsymbol{n}\right) \\
&= \mathbf{1}\left(\boldsymbol{a}_{\alpha,\beta} \cdot \boldsymbol{n}\right) =: b_{\alpha\beta}
\end{aligned} \tag{39}$$